



Course Name

ELEMENTARY MATHEMATICS II

FULL COURSE SUMMARY

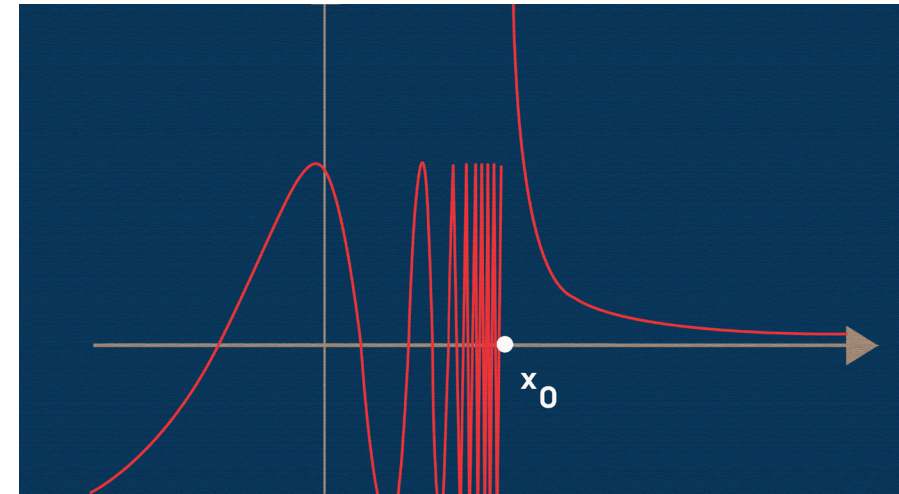
IN SUMMARY

Firstly, you learned about the various aspects of mathematical analysis. Real-valued functions are mathematical constructs that assign real numbers to their inputs. This topic involves understanding key concepts and properties associated with these functions. You also learned that a real-valued function is a rule that associates each element from a set (usually real numbers) with a unique real number. This forms the basis for further exploration. Additionally, you learned about the domain and range of a function, graphs of a function, properties of a function, types of a function, operations on functions, and some function applications.

Next, you learned about different types of functions, which are the fundamental building blocks of mathematical relationships. Functions act as transformation machines, taking an input and producing a unique output, much like a recipe. We explored various types of functions, including linear, quadratic,

exponential, logarithmic, trigonometric, piecewise, absolute value, polynomial, and constant functions. Each type has unique characteristics, and understanding these functions is crucial for graphing and analysing them. Furthermore, you learned about the fundamental concept of limits in calculus and how they are expressed using mathematical notation. Limits represent the value that a function approaches as the input, typically denoted as ' x ,' gets closer to a specific point, usually denoted as a . You also learned about the behaviour of functions as they approach particular points or infinity. We also discussed how to evaluate limits algebraically and the significance of limits in various applications, including real-world scenarios.

Lastly, you learned about the fundamental concepts of continuity and discontinuity of limits. Continuity in a function refers to its smooth, uninterrupted behaviour, where you can draw its graph at a specific point without lifting your pen. This means that the limit of the function at that point must exist and match the function's value. On the other hand, discontinuity occurs when a function exhibits abrupt breaks, jumps, or holes in its graph at specific points. Discontinuities include removable (or holes), jump, infinite, and essential. You also learned some practical applications of these concepts in various fields, such as engineering, physics, economics, and computer science.



BULLET POINT SUMMARY

- Real-valued functions assign real numbers to their inputs and play a fundamental role in mathematics and its applications.
- Real-valued functions have defined domains (input values) and ranges (output values), crucial for understanding their behaviour.
- Functions represent mathematical relationships.
- Limits represent the value that a function approaches as the input gets closer to a specific point.

Key references:

Stewart, J. (2015). Calculus: Early Transcendentals. Cengage Learning.
Anton, H., Bivens, I., & Davis, S. (2016). Calculus. Wiley.

- Limits are crucial in calculus, enabling the study of rates of change, and accumulation, and have applications in numerical analysis, animation, and game development.
- A function is continuous when there are no abrupt breaks or jumps in its graph at a specific point. The limit and the function's value at that point must match for continuity to exist.
- Discontinuity occurs when a function exhibits abrupt breaks, jumps, or holes in its graph at specific points.

CASE STUDY

CASE STUDY 1:

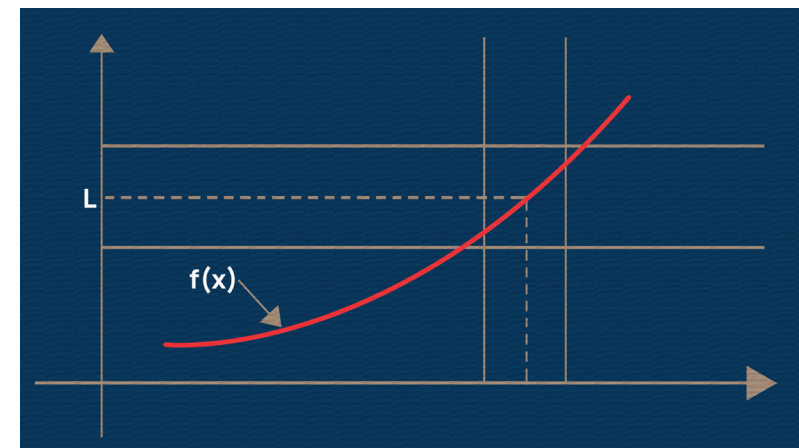
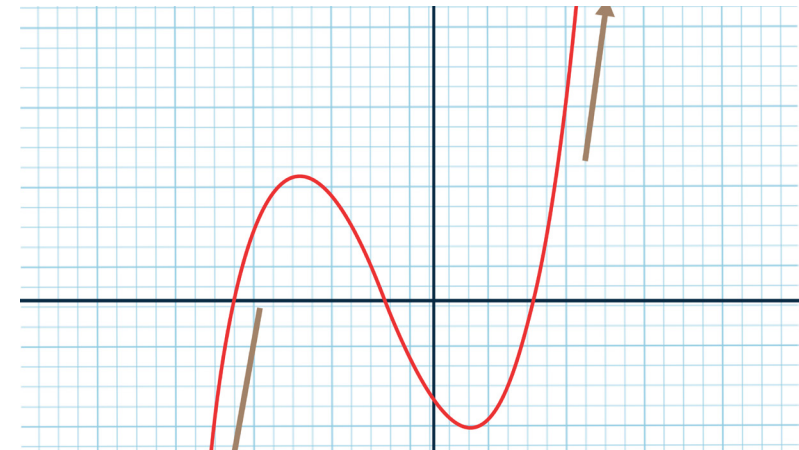
TITLE: EVALUATING LIMITS IN STRUCTURAL ENGINEERING

The case study demonstrates the practical application of evaluating limits algebraically in structural engineering, specifically focusing on determining the maximum load a bridge can bear before structural failure occurs. Through a real-world engineering project, the study outlines the methodology of gathering data, applying algebraic methods to assess limits, comparing calculated limits with actual load-bearing capacity, and analysing results to evaluate the accuracy and safety of the structural design.

CASE STUDY 2:

TITLE: GRAPHING FUNCTIONS IN REAL-WORLD APPLICATIONS

This case study demonstrates the practical application of graphing functions in engineering design, emphasising their role in optimising system parameters and enhancing efficiency. By identifying the appropriate function type and utilising graphing techniques, engineers can analyse and optimise engineering systems, leading to more effective and efficient designs.



Key references:

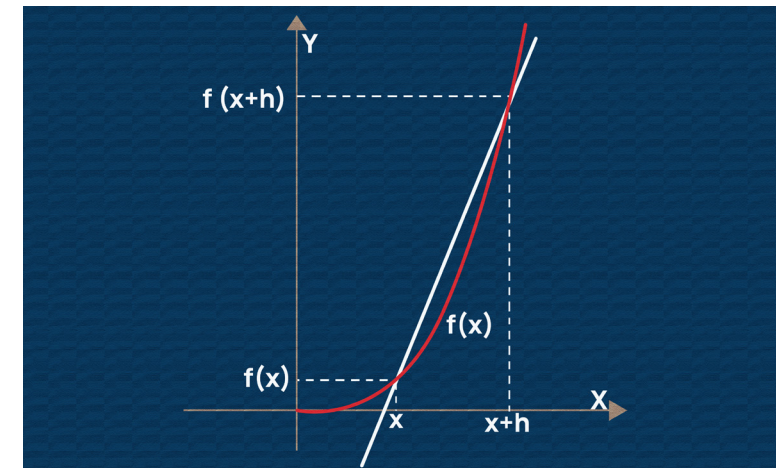
Larson, R., & Edwards, B. H. (2017). Calculus of a Single Variable. Cengage Learning.
Simmons, G. F. (2007). Calculus with Analytic Geometry. McGraw-Hill Education.

QUESTIONS TO PONDER

- How can the integration of real-valued functions in engineering design be further optimised to enhance the efficiency and effectiveness of complex system analysis?
- What are the challenges and limitations of using limits in structural design, and how can they be overcome?
- How can engineers optimise the use of graphical techniques for evaluating limits in complex system designs to enhance efficiency and effectiveness?

SKILLS AND COMPETENCIES YOU HAVE ACQUIRED AFTER THIS LESSON

Understanding real-valued functions, Mathematical proficiency, Problem-solving, Critical thinking, Engineering application, Analytical thinking, Communication, and Understanding limits.



PERSONAL REFLECTION

The lesson on graphical techniques and continuity concepts reinforced mathematics' pivotal role in diverse real-world problem-solving scenarios, emphasising its practical relevance across scientific fields.

Key references:

"Calculus Know-It-ALL" by Stan Gibilisco.

"Theory and Problems of Advanced Calculus" by Robert Wrede, and Murray R. Spiegel.

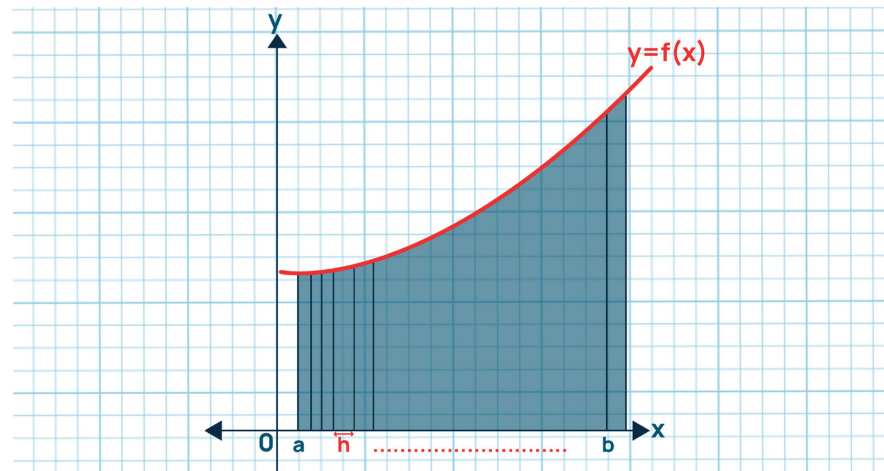
IN SUMMARY

Firstly, you learned the fundamental concepts of differentiation in calculus. The derivative provides the rate at which a function changes with respect to its independent variable. Differentiation involves finding the derivative, which can be denoted as f' , dy/dx , or y' . We discussed various orders of derivatives, including first, second, and higher-order derivatives, and the notations used to represent them. Additionally, we covered different methods of finding derivatives, such as the derivative from the first principle, the power rule, the product rule, the quotient rule, the chain rule, and the derivative of implicit functions. Next, you learned about higher-order derivatives in calculus. Higher-order derivatives are mathematical concepts that allow us to understand how the rate of change of a function changes. They are denoted as the function raised to the power of the n th term, where n is a non-negative integer. Higher-order derivatives can be calculated using the rules of differentiation, such as the power rule,

product rule, quotient rule, and chain rule. Higher-order derivatives are important in many fields, such as physics, engineering, and economics. Which are used to understand motion, and forces, analyse circuits, and control systems, model economic variables, and provide deeper insights into a function's behaviour. Furthermore, you learned about the fundamental concept of integration in calculus. Integration is a mathematical technique that allows us to find the accumulation of quantities. It involves calculating the area under a curve or determining the change in a quantity. There are two main types of integrals: definite and indefinite. Definite integrals are used to find the exact value of the accumulated quantity between specific points on a graph, while indefinite integrals help us find a general formula for the accumulated quantity without specifying limits. The integration process results in an integral, and understanding these integrals is crucial for solving various mathematical problems. Lastly, you learned about integration techniques, a fundamental cornerstone of calculus. The lesson covered various methods of integration and the techniques required to apply them effectively. These techniques include integration by substitution, integration by parts, and partial fraction decomposition. Through comprehensive examples, you gained a deeper understanding of how to apply these methods to solve complex integrals.

BULLET POINT SUMMARY

- Methods of finding derivatives include the first principle, power rule, product rule, quotient rule, chain rule, and derivative of implicit functions.
- Applications of derivatives include critical point identification, concavity, increasing and decreasing functions, maxima and minima.
- Higher-order derivatives are derivatives of derivatives, representing how the rate of change changes. They are denoted as functions raised to the power of the n th term, and n can be any non-negative integer.



Key references:

Larson, R., & Edwards, B. H. (2017). *Calculus of a Single Variable*. Cengage Learning.
Simmons, G. F. (2007). *Calculus with Analytic Geometry*. McGraw-Hill Education.

- Various orders of derivatives, from the first to higher orders, have distinct purposes and applications.
- Notations like prime notation and Leibniz's fractional notation are used to represent these derivatives.
- Definite integrals are used to calculate the exact accumulated quantity between specific points, and indefinite, provide a general formula for accumulated quantities.
- Integration is the process of finding the antiderivative of a function, which is the function whose derivative is the original function.

CASE STUDY

CASE STUDY 1:

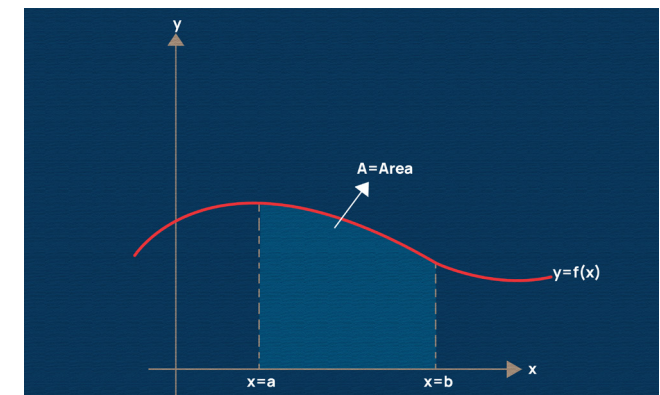
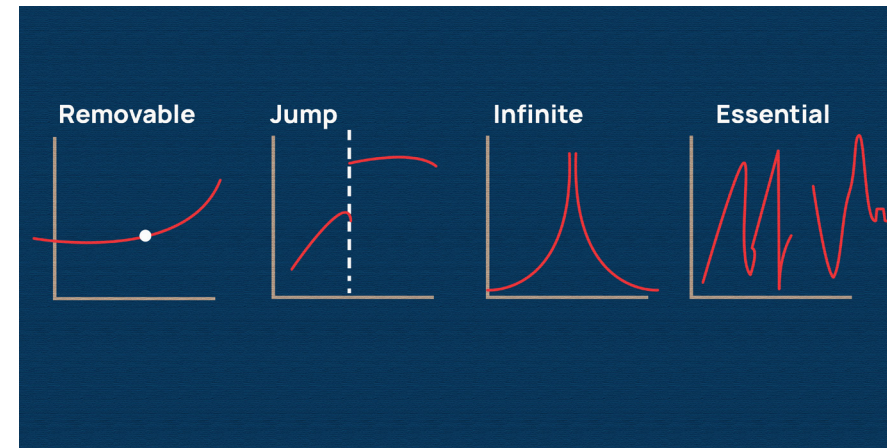
TITLE: CALCULATING DAILY SALES USING INTEGRATION

This case study explores the application of integration in solving a real-world problem – calculating daily sales. Integration, a fundamental calculus concept, aids in finding the area under the curve representing sales. The methodology involves identifying the sales function, finding its antiderivative, and evaluating it over the desired time interval. The results will demonstrate how integration concepts can effectively solve practical problems in undergraduate studies, showcasing their relevance in real-world applications.

CASE STUDY 2:

TITLE: APPLICATION OF INTEGRATION IN STRUCTURAL ENGINEERING

This case study investigates the application of higher-order derivatives in engineering design, specifically in the control and optimization of complex systems. Focused on a real-world example in control systems engineering, the study aims to collect and analyse data from the design of an automated control system. The objective is to understand how higher-order derivatives contribute to achieving system optimisation and control, shedding light on their pivotal role in engineering, particularly in complex system design and stability.



Key references:

Larson, R., & Edwards, B. H. (2017). Calculus of a Single Variable. Cengage Learning.
Simmons, G. F. (2007). Calculus with Analytic Geometry. McGraw-Hill Education.

QUESTIONS TO PONDER

- How can the integration of derivative concepts in structural engineering be further optimised to enhance the efficiency and effectiveness of complex structural analysis and design?
- How can the integration of higher-order derivatives in engineering design be further optimised to enhance the efficiency and effectiveness of complex system control?
- In what ways can engineering education better emphasise the practical applications of integration to prepare future structural engineers for real-world problem-solving challenges?

SKILLS AND COMPETENCIES YOU HAVE ACQUIRED AFTER THIS LESSON

Understanding of differentiation, mathematical proficiency, problem-solving, critical thinking, and its application in engineering and various fields. Ability to interpret functions geometrically and analyse real-world applications.

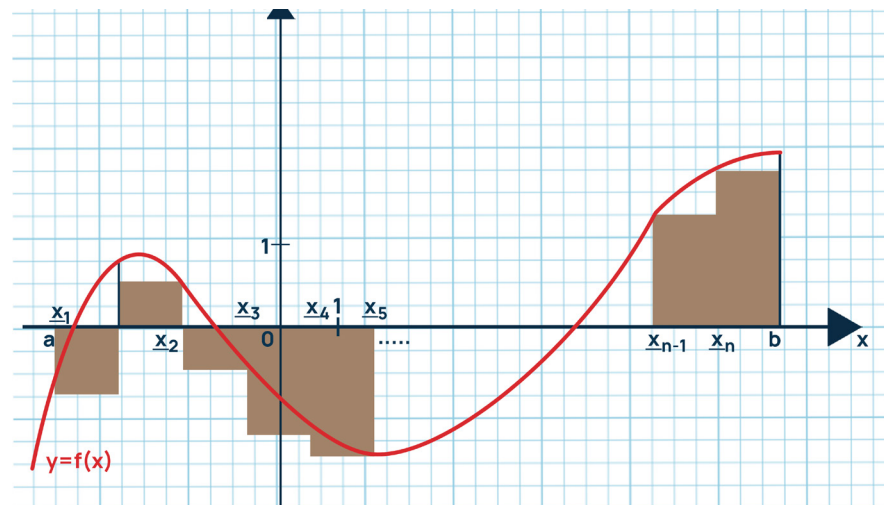
PERSONAL REFLECTION

Studying higher-order derivatives has deepened my understanding of how quantities change and the changes in this change. It has also equipped me with the ability to tackle intricate mathematical problems and real-world applications in fields like Engineering, Physics, and Economics.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

IN SUMMARY

Firstly, you learned about improper integrals, their convergence and divergence, and the Comparison test for improper integrals. Improper integrals are integrals with infinite intervals or discontinuities. They can either converge to a finite value or diverge to infinity. The Comparison test is a powerful tool that allows us to determine the convergence or divergence of complex improper integrals by comparing them to simpler, known integrals. Then you learned that definite integrals enable us to determine the exact area bounded by functions and the x-axis. The key concept is that the integral represents the signed area enclosed between the curve and the x-axis from a starting point 'a' to an endpoint 'b'. We achieved this by breaking down the region into infinitesimally small vertical strips, meticulously calculating each strip's area, and then summing them up. This process provides the net area between the curve and the x-axis. You also learned the concept of Riemann sums, which allow us to calculate the area beneath complex curves. Additionally, you learned how definite integrals can be applied to find the area between



two curves and dealt with examples where the curves intersect or have intricate features. Definite integrals prove to be a powerful tool for precisely determining areas between curves, regardless of their complexity. Subsequently, you learned about the application of definite integrals to find the volumes of solids of revolution. You learned how to use the disk method, washer method, and shell method to calculate the volumes of a variety of three-dimensional objects. You also learned about some of the real-world applications in engineering, physics, and architecture. Lastly, you learned how integral calculus is applied to solve real-life problems in various fields, including physics, engineering, economics, and environmental science. Practical applications of integration in these domains, showcasing how mathematical concepts can be used to address complex real-world challenges were taught. We also explored how integration is used to relate velocity and acceleration in physics, calculate work done by varying forces, analyse electric and magnetic fields in engineering, and much more. Integration was also shown to be a valuable tool in economics for analysing changes in cost, revenue, and profit.

BULLET POINT SUMMARY

- The comparison test is employed in assessing the convergence or divergence of improper integrals by comparing them to known integrals, where the dominating integral dictates the behaviour of the smaller one.
- Definite integrals can be used to calculate the area under a curve, the area between two curves, and the area within polar coordinates.
- Definite integrals can also be used to determine the area of three-dimensional curves and the area of irregular shapes.
- Definite integrals are a fundamental concept in calculus with countless practical implications in fields spanning from physics to engineering and economics.

- There are two common categories of solids of revolution: those formed by rotating a curve around the x-axis and those formed by rotating a region enclosed by two curves around the x-axis.
- The disk method, shell method, and washer method are three common methods for calculating the volumes of solids of revolution.
- Integration's role in engineering includes such things as analysing electric and magnetic fields, determining the distribution of forces within structures, and optimising fluid systems.

CASE STUDY

CASE STUDY 1:

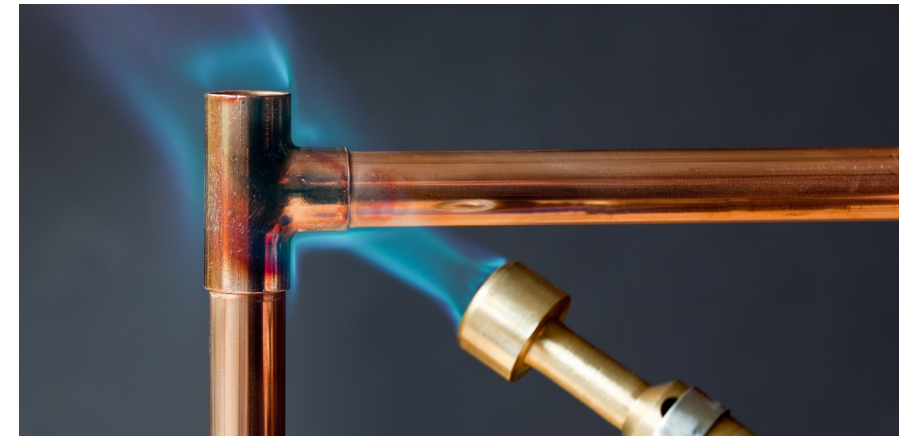
TITLE: ANALYSING TEMPERATURE CONDUCTION IN MATERIALS

In this case study, the focus is on analysing temperature conduction in materials with implications for industrial processes. The objective is to investigate the convergence or divergence of improper integrals to understand heat dissipation over time. Using the comparison test on materials with different thermal conductivity, the study aims to determine if heat dissipation rates converge or diverge, offering insights into material behaviour crucial for industrial applications. The results will inform material selection, impacting stability in heat transfer for various industrial processes.

CASE STUDY 2:

TITLE: USING DEFINITE INTEGRALS TO CALCULATE THE CROSS-SECTIONAL AREA OF A DAM

This case study explores the practical application of definite integrals, focusing on calculating the cross-sectional area of a dam—a crucial factor in assessing its strength and stability. The methodology involves identifying the dam's cross-sectional profile function, determining the calculation interval, evaluating the definite integral, and interpreting the result as the area. The study showcases the versatility of definite integrals in diverse industries, with results contingent on the unique characteristics of the addressed dam.



Key references:

Larson, R., & Edwards, B. H. (2017). Calculus of a Single Variable. Cengage Learning.
Simmons, G. F. (2007). Calculus with Analytic Geometry. McGraw-Hill Education.

QUESTIONS TO PONDER

- How can the knowledge of improper integrals and the comparison test be applied to optimise heat conduction processes in real-world industrial applications?
- How can definite integrals be used to solve real-world problems?
- What are the different methods for calculating the volume of solids of revolution?
- How can integration be further optimised in environmental impact assessments to enhance accuracy and effectiveness in evaluating pollutant effects?

SKILLS AND COMPETENCIES YOU HAVE ACQUIRED AFTER THIS LESSON

Mathematical analysis, Problem-solving, Critical thinking, Proficiency in applying definite integrals to calculate land areas, Mathematical modelling, Ability to quantify and assess complex real-world issues.

PERSONAL REFLECTION

This lesson deepened my understanding of the importance of improper integrals in real-world applications. It highlighted the practical utility of mathematical concepts in solving complex industrial problems and how a strong foundation in calculus is crucial for scientific and engineering endeavours.



Key references:

"Calculus Know-It-ALL" by Stan Gibilisco.

"Theory and Problems of Advanced Calculus" by Robert Wrede, and Murray R. Spiegel.